Descriptive Summary of the Data

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.3	66.33	66.37	72.1	62.28	288655.41
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Min	1	40.89	37.0	50.0	50.0	51.21	200000.0
Q1-25th%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Median / Q2-50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3-75th%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99th%	212.86	87.0	91.86	83.86	97.0	76.11	NaN
Max-100th%	215	89.4	97.7	91.0	98.0	77.89	940000.0
IQR	107.0	15.1	12.1	11.0	23.5	8.31	60000.0
1.5rule	160.5	22.65	18.15	16.5	35.25	12.465	90000.0
LesserRange	-106.0	37.95	42.75	44.5	24.75	45.48	150000.0
GreaterRange	322.0	98.35	91.15	88.5	118.75	78.72	390000.0
LesserOutlier	False	False	True	False	False	False	False
GreaterOutlier	False	False	True	True	False	False	True

The given Placement dataset contains students' or employees' information with the following columns:

sl_no – Serial number or ID
ssc_p – 10th grade (SSC) percentage
hsc_p – 12th grade (HSC) percentage
degree_p – Graduation percentage
etest_p – Employability test percentage
mba_p – MBA percentage
salary – Salary offered

The above table shows:

Mean	- average
Median / Q2	- middle value
Mode	- most frequent value
Min and Max	- range of data
Q1, Q3	- Quartiles which divide data into 25% and 75% groups
99th % Percentile	- To see the extreme high values
IQR	- Interquartile Range = Q3 – Q1, showing the spread of the middle 50% of the data
1.5 Rule	- used to detect outliers
LesserRange GreaterRange	- indicate the possible range for detecting outliers

Key Observations

1. SSC (10th Grade) Scores

a) Scores range 40.89% to 89.4%, median 67%; most students are first-class, and 75% scored above 60.6%, showing strong performance with almost no extreme outliers.

b) Middle 50% of students scored between 60.6% and 75.7% (IQR 15.1%), meaning lower scorers are not far (7%) from the median, and top performers are slightly spread out (8%).

c) Performance is consistent, with the majority scoring well and no significant low or high deviations.

2. HSC (12th Grade) Scores

a) Scores range 37% to 97.7%, median 65%, and 75% of students scored above 60.9%, showing overall strong academics with one low and one high outlier.

b) Middle 50% scored between 60.9% and 73% (IQR 12.1%), meaning the lower half is tightly packed (4%), and top students are a bit more spread out (8%).

c) Distribution is wider than SSC, indicating some variation at extremes - Both Lesser and Greater Range Outliers, but most students still perform well above average.

3. Degree Scores

a) Scores range 50% to 91%, median 66%, and 75% of students scored above 61%, showing consistent graduation performance.

b) Middle 50% scored between 61% and 72% (IQR 11%), meaning most students are very close to each other in scores (5%-6%), and only mild positive outliers exist.

c) Performance is very stable, with no low outliers and only few high achievers above 88.5% (Greater Range Outlier).

4. MBA Scores

a) Scores range 51.21% to 77.89%, median 62%, and 75% scored above 57.95%, slightly lower than degree scores but still consistent.

b) Middle 50% scored between 57.95% and 66.25% (IQR 8.31%), which is the tightest spread (~4%), meaning almost all students perform similarly.

c) MBA scores are the most uniform, with no major outliers and very few variations across students.

5. Employability Test (E-Test)

a) Scores range 50% to 98%, median 71%, and 75% scored above 60%, showing students perform better in aptitude than academics.

b) Middle 50% scored between 60% and 83.5% (IQR 23.5%), the widest spread (11% - 12.5%), meaning student aptitude varies widely with a few exceptional high performers.

c) Top scorers (above 83.5%) are clear positive outliers, indicating few students excel significantly in employability tests.

6. Salary

a) Salaries range 2,00,000 to 9,40,000, median 2.65 LPA, and most earn between 2.4–3 LPA, with very few students getting very high packages.

b) Middle 50% salaries are 2.4–3 LPA (IQR 60,000), meaning most salaries are clustered in a narrow range (25,000 - 35,000 difference).

c) Salary distribution is right-skewed, where few top earners (above 3.9 LPA) drive the maximum to 9.4 LPA, highlighting Greater Range Outlier.

Why do we use 1.5 × IQR Rule for Outlier Detection?

The IQR method is a common way to find outliers in a dataset. It looks at the range where the middle 50% of data lies, called the Interquartile Range (IQR). If a value is 1.5 times the IQR below the first quartile (Q1) or above the third quartile (Q3), it is considered an outlier. This 1.5 factor is used because it works well for data that follows a normal (bell-shaped) distribution.

The normal or Gaussian distribution is important because many real-life data points naturally follow this bell curve, like heights, weights, and exam scores. It is also the basis for many statistical calculations and predictions, thanks to the Central Limit Theorem, which explains why averages of random samples tend to form a bell-shaped curve.